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ADMS

16.25.1

Reports User Guide

Copyright

Change History

Date	Change Description
Jan 26, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Apr 16, 2024	ADMS 16.7.0: • Updated document format and styling.

Contents

1. Introduction

The ADMS Reports application provides some sample distribution and outage management reports that power companies may use directly, or as templates for their own custom reports.

The data source for these reports is the Archive database, which contains historical data such as incidents, switching orders, safety documents, Distribution Power Flow (DPF) summaries, and Distribution Network Analysis Functions (DNAF) solutions. For a complete list of the types of data that may be stored in the Archive database, refer to the *ADMS Archive User Guide*.

The reports are accessible from any web browser. Users may specify the desired date range and export the reports in various formats.

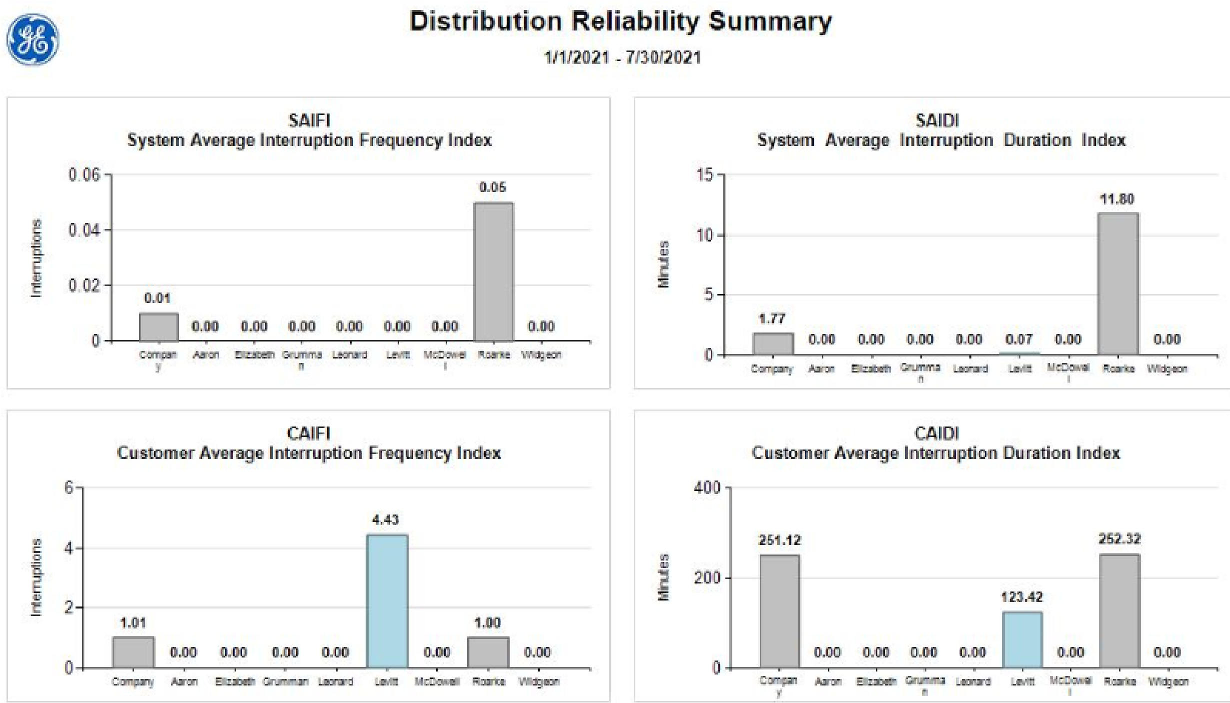


Figure 1. Sample Summary Report

2. Viewing Reports

This chapter provides guidance on running the ADMS Reports application and viewing reports.

2.1. Accessing the Reports

The ADMS Reports are based on the SQL Server Reporting Services (SSRS). For more information about the SSRS, see https://docs.microsoft.com/en-us/sql/reporting-services/create-deploy-and-manage-mobile-and-paginated-reports?view=sql-server-ver15.

To access the Reports page:

- **For Oracle:** From the Start menu, navigate to `All Programs> Eterra> Distribution> Tools> Sample Reports Oracle`.
- **For MS SQL Server:** From the Start menu, navigate to `All Programs> Eterra> Distribution> Tools> Sample Reports Microsoft SQL Server`.

Alternatively, in a web browser open one of these URLs:

- **For Oracle:** http://localhost/Reports/Pages/Folder.aspx?ItemPath=%2fADMS_Reports_ORA
- **For MS SQL Server:** http://localhost/Reports/Pages/Folder.aspx?ItemPath=%2fADMS_Reports_MSSQL

To access reports on another report server, replace "localhost" with the name or IP address of the remote report server.

The Reports page is loaded at startup, as shown in the following figure.

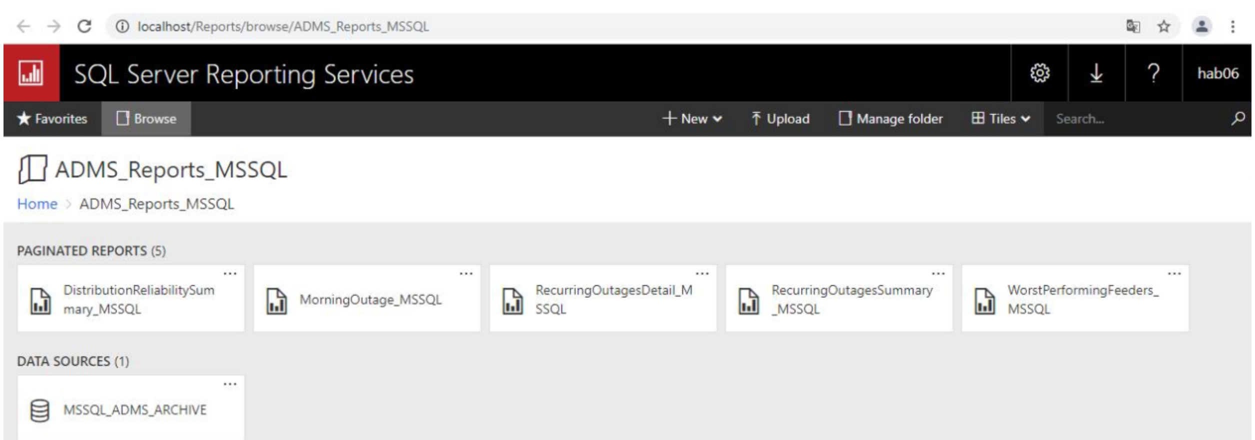


Figure 2. Reports Page

> **Note**
>
> The Reports may take a moment to load.

The Reports page contains all available reports and a shared data source (for example, ORA_ADMS_ARCHIVE) where the database connection parameters are stored. This shared data source is automatically created during installation.

2.2. Reports Toolbar

The Reports toolbar is available for each of the ADMS Reports.

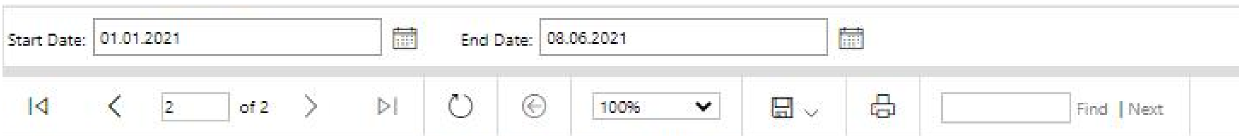


Figure 3. Reports Toolbar

Using the Reports toolbar, you can:

- Select the reporting period
- Navigate between pages
- Select the zoom level
- Search for text
- Export to files (XML, PDF, MHTML, Excel, TIFF, Word)
- Refresh the page
- Print the report
- Export to a data feed

2.3. Reports Overview

This section provides an overview of current sample reports.

As ADMS Reports includes the same reports for Archive databases on either Oracle or SQL Server, this guide does not provide separate descriptions for these databases.

Each report has Start/End Date fields, which can be used to specify the reporting period. To generate a report with new settings, click the View Report button.

2.3.1. Distribution Reliability Summary

The Distribution Reliability Summary report shows the distribution reliability indices for the company and all cities.

By default, the report is generated for the year to date.

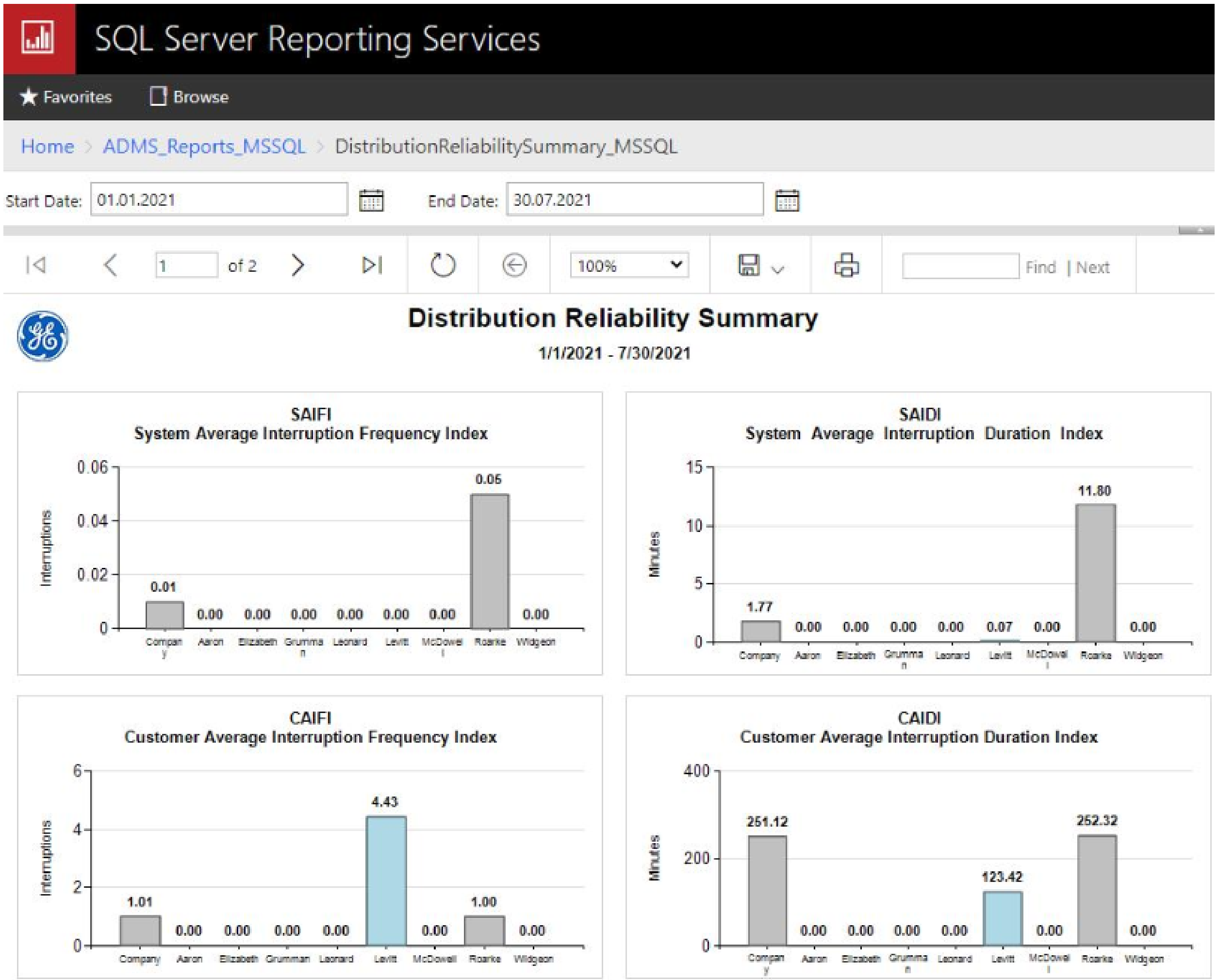


Figure 4. Distribution Reliability Summary

The following information about reliability performance indices was taken from the IEEE Std 1366-2012 *Guide for Electric Power Distribution Reliability Indices*.

SAIFI – System Average Interruption Frequency Index

Indicates how often the average customer experiences a sustained interruption (incident) over a predefined period of time.

$$\mathbf{SAIFI} = \frac{\sum \textbf{Total Number of Customer Interruptions}}{\textbf{Total Number of Customers Served}}$$

SAIDI – System Average Interruption Duration Index

Indicates the total duration of interruption (incident) for the average customer during a predefined period of time.

$$\mathrm{SAIDI} = \frac{\sum \text{Customer Minutes of Interruption (CMI)}}{\text{Total Number of Customers Served}}$$

$$\mathrm{CMI} = \sum (\text{Number of Interrupted Customers} \times \text{Interruption Duration})$$

CAIFI – Customer Average Interruption Frequency Index

Represents the average frequency of sustained interruptions (incidents) for those customers experiencing sustained interruptions. The customer is counted once, regardless of the number of times interrupted for this calculation.

$$\mathrm{CAIFI} = \frac{\sum \text{Total Number of Customer Interruptions}}{\text{Total Number of Distinct Customers Interrupted}}$$

CAIDI – Customer Average Interruption Duration Index

Represents the average time that was required to restore service.

$$\mathrm{CAIDI} = \frac{\sum \text{Customer Minutes of Interruption (CMI)}}{\text{Total Number of Customers Interrupted}}$$

MAIFI – Momentary Average Interruption Frequency Index

Indicates the average frequency of momentary interruptions (incidents).

$$\mathrm{MAIFI} = \frac{\sum \text{Total Number of Customer Momentary Interruptions}}{\text{Total Number of Customers Served}}$$

MAIFI_E – Momentary Average Interruption Event Frequency Index

Indicates the average frequency of momentary interruption events (incidents). This index does not include the events immediately preceding a sustained interruption.

$$\mathrm{MAIFI}_E = \frac{\sum \text{Total Number of Customer Momentary Interruption Events}}{\text{Total Number of Customers Served}}$$

ASAI – Average Service Availability Index

Represents the fraction of time (often in percentage) that a customer has received power during the defined reporting period.

$$\mathrm{ASAI} = \frac{\text{Customer Hours Service Availability}}{\text{Customer Hours Service Demand}}$$

$$\text{ASAI} = \frac{\text{Total Number of Customers Served} \times \text{Report Time Period} - \text{CMI}}{\text{Total Number of Customers Served} \times \text{Report Time Period}}$$

CTAIDI – Customer Total Average Interruption Duration Index

Represents the total time in the reporting period that average customers who actually experienced an interruption were without power. This index is a hybrid of CAIDI and is similarly calculated, except that those customers with multiple interruptions are counted only once.

$$\text{CTAIDI} = \frac{\text{CMI}}{\text{Total Number of Distinct Customers Interrupted}}$$

$CEMI_5$ – Customers Experiencing Multiple Interruptions Index

Indicates the ratio of individual customers experiencing 5 or more sustained (non-momentary) interruptions to the total number of customers served.

$$\text{CEMI}_5 = \frac{\text{Total Number of Customers Experienced 5 or More Sustained Interruptions}}{\text{Total Number of Customers Served}}$$

$CEMSMI_5$ – Customers Experiencing Multiple Sustained Interruption and Momentary Interruption Events Index

Indicates the ratio of individual customers experiencing one or more of both sustained interruptions (incidents) and momentary interruption events to the total customers served. Its purpose is to help identify customer issues that can't be observed by using averages.

$$\text{CEMSMI}_5 = \frac{\text{Total Number of Customers Experiencing 5 or More Interruptions}}{\text{Total Number of Customers Served}}$$

CELID-4 – Customers Experiencing Long Interruption Durations Index

Indicates the ratio of individual customers that experience interruptions with durations longer than or equal to a given time (4 hours). That time is either the duration of a single interruption or the total amount of time that a customer has been interrupted during the reporting period.

$$\text{CELID-4} = \frac{\text{Total Number of Customers that Experienced 4 or More Hours Duration}}{\text{Total Number of Customers Served}}$$

CELID-6 – Customers Experiencing Long Interruption Durations Index

Indicates the ratio of individual customers that experience interruptions with durations longer than or equal to a given time (6 hours). That time is either the duration of a single interruption or the total amount of time that a customer has been interrupted during the reporting period.

$$\text{CELID}-6=\frac{\text{Total Number of Customers that Experienced 6 or More Hours Duration}}{\text{Total Number of Customers Served}}$$

2.3.2. Morning Outage

The Morning Outage report shows outages by city. The data is sorted by incident number in ascending order.

By default, the report is generated for the period from 6:00 AM of the previous day to 6:00 AM of the current day.

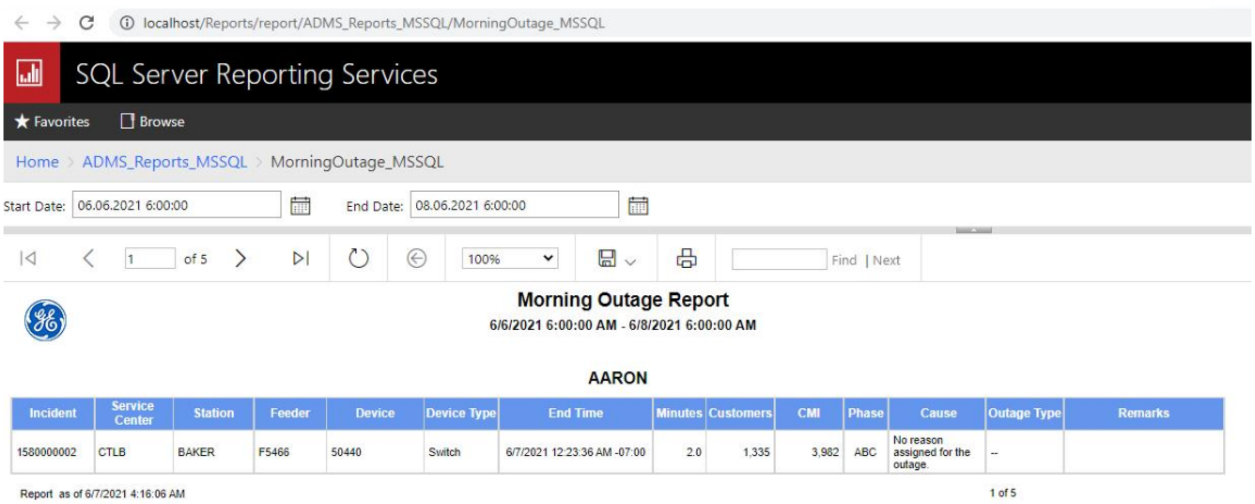


Figure 5. Morning Outage Report

The Morning Outage Report table includes the following columns:

- **Incident:** The incident's order number.
- **Service Center:** Service center's name.
- **Station:** Substation's name.
- **Feeder:** Feeder's name.
- **Device:** Device's name.
- **Device Type:** Type of the device.
- **End Time:** Incident's end time.
- **Minutes:** Incident's duration in minutes.
- **Customers:** The number of affected customers.
- **CMI:** Customer Minutes Interrupted.
- **Phase:** Phases affected by the incident.
- **Cause:** Cause of the incident.
-

Outage Type: Incident's outage type.

- **Remarks:** Incident's comments.

2.3.3. Recurring Outages Summary

The Recurring Outages Summary report shows the number of devices that are the device of record for multiple outages. The reports shows data for the company and all cities.

By default, the report is generated for the year-to-date and counts only devices that are the device of record for 4 or more outages.

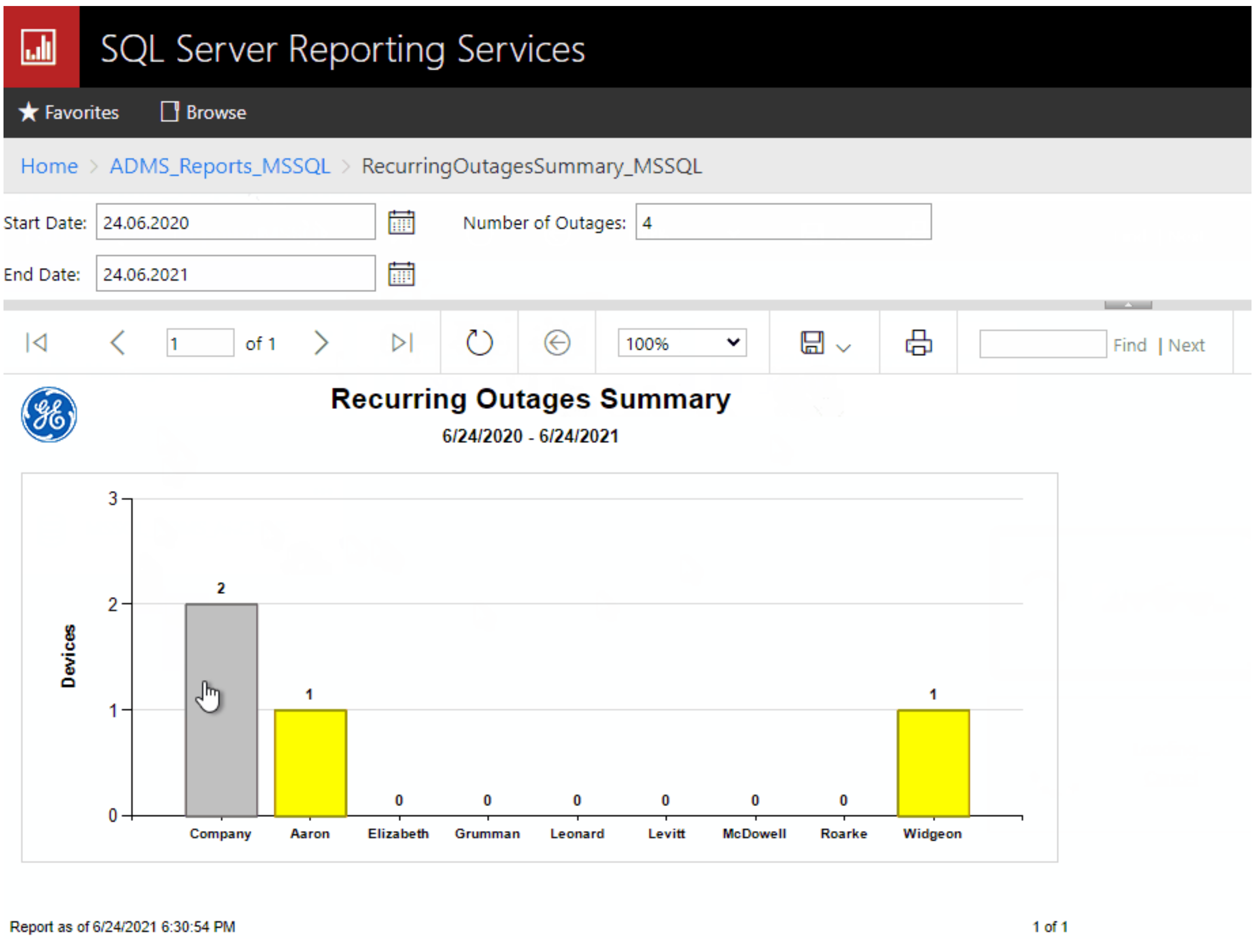


Figure 6. Recurring Outages Summary

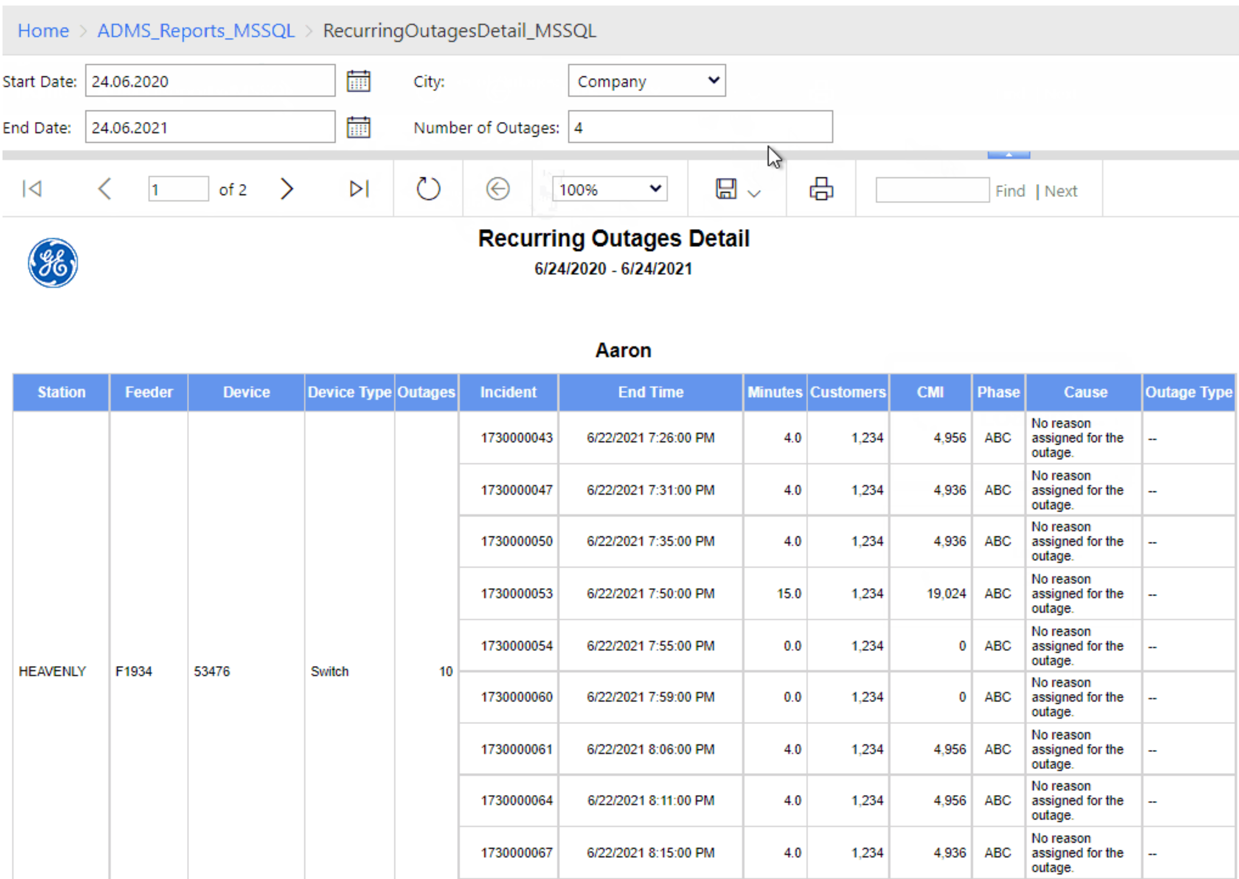
Clicking on a rectangle navigates to the Recurring Outages Detail report containing the relevant details.

2.3.4. Recurring Outages Detail

The Recurring Outages Detail report shows the devices that are the device of record for multiple outages.

By default, the report is generated for the yea-to-date, for all cities, and includes only devices that are the device of record for 4 or more outages. The reports can also show data for the company or city selected on the Recurring Outages Summary report.

For reports for the company, devices are sorted by city, service center, station, and feeder. For reports for a service center, devices are sorted by station, and feeder.


Home > ADMS_Reports_MSSQL > RecurringOutagesDetail_MSSQL

Start Date: 24.06.2020 City: Company

End Date: 24.06.2021 Number of Outages: 4

1 of 2 100% Find | Next

Recurring Outages Detail

6/24/2020 - 6/24/2021

Aaron

Station	Feeder	Device	Device Type	Outages	Incident	End Time	Minutes	Customers	CMI	Phase	Cause	Outage Type
HEAVENLY	F1934	53476	Switch	10	1730000043	6/22/2021 7:26:00 PM	4.0	1,234	4,956	ABC	No reason assigned for the outage.	--
					1730000047	6/22/2021 7:31:00 PM	4.0	1,234	4,936	ABC	No reason assigned for the outage.	--
					1730000050	6/22/2021 7:35:00 PM	4.0	1,234	4,936	ABC	No reason assigned for the outage.	--
					1730000053	6/22/2021 7:50:00 PM	15.0	1,234	19,024	ABC	No reason assigned for the outage.	--
					1730000054	6/22/2021 7:55:00 PM	0.0	1,234	0	ABC	No reason assigned for the outage.	--
					1730000060	6/22/2021 7:59:00 PM	0.0	1,234	0	ABC	No reason assigned for the outage.	--
					1730000061	6/22/2021 8:06:00 PM	4.0	1,234	4,956	ABC	No reason assigned for the outage.	--
					1730000064	6/22/2021 8:11:00 PM	4.0	1,234	4,956	ABC	No reason assigned for the outage.	--
					1730000067	6/22/2021 8:15:00 PM	4.0	1,234	4,936	ABC	No reason assigned for the outage.	--

Figure 7. Recurring Outages Detail

The Recurring Outages Detail table includes the following columns:

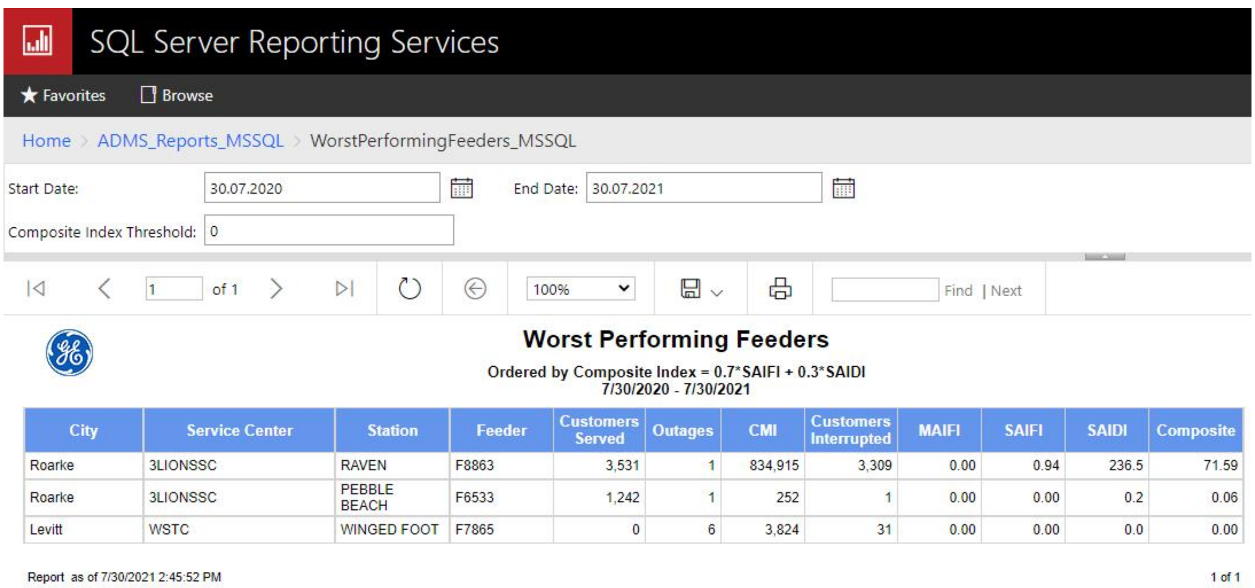
- **Station:** Substation's name.
- **Feeder:** Feeder's name.
- **Device:** Device's name.
- **Device Type:** Type of the device.
- **Outages:** The number of incidents for the device.
- **Incident:** The incident's order number.
- **End Time:** Incident's end time.
- **Minutes:** Incident's duration in minutes.

- **Customers:** The number of affected customers.
- **CMI:** Customer Minutes Interrupted.
- **Phase:** Phases affected by the incident.
- **Cause:** Cause of the incident.
- **Outage Type:** Incident's outage type.

2.3.5. Worst Performing Feeders

The Worst Performing Feeders report shows feeders whose Composite Index exceeds a specified threshold. Composite Index is equal to (0.7*SAIFI) + (0.3*SAIDI). The data is sorted by the Composite Index value in descending order.

By default, the report is generated for the year-to-date, with a Composite Index threshold of 0 (so it includes all feeders).



City	Service Center	Station	Feeder	Customers Served	Outages	CMI	Customers Interrupted	MAIFI	SAIFI	SAIDI	Composite
Roarke	3LIONSSC	RAVEN	F8863	3,531	1	834,915	3,309	0.00	0.94	236.5	71.59
Roarke	3LIONSSC	PEBBLE BEACH	F6533	1,242	1	252	1	0.00	0.00	0.2	0.06
Levitt	WSTC	WINGED FOOT	F7865	0	6	3,824	31	0.00	0.00	0.0	0.00

Report as of 7/30/2021 2:45:52 PM 1 of 1

Figure 8. Worst Performing Feeders

The Worst Performing Feeders table includes the following columns:

- **City:** City's name.
- **Service Center:** Service center's name.
- **Station:** Substation's name.
- **Feeder:** Feeder's name.
- **Customers Served:** The number of customers supplied by the feeder.
- **Outages:** The number of incidents for the device.
- **CMI:** Customer Minutes Interrupted.
-

Customers Interrupted: The number of customers who experienced service interruption.

- **MAIFI:** Momentary Average Interruption Frequency Index. The average frequency of momentary interruptions.
- **SAIDI:** System Average Interruption Duration Index. Indicates the total duration of interruption for the average customer during a predefined period of time.
- **SAIFI:** System Average Interruption Frequency Index. Indicates how often the average customer experiences a sustained interruption over a predefined period of time.
- **Composite:** The composite index value for the feeder. Calculates as (0.7*SAIFI) + (0.3*SAIDI).